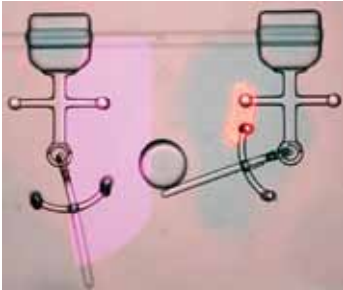




3D printing enables autonomous functions for microrobots

Designing movable actuators driven by chemical energy has been a difficult and time-consuming process. Researchers at the Department of Mechanical Engineering at Osaka University have developed a method for manufacturing



3D-printing microrobots (Credit: Keisuke Morishima, Science Robotics (2022)). DOI: 10.1126/scirobotics.aba8212)

complex microrobots driven by chemical energy using in-situ integration. By 3D printing and assembling the mechanical structures and actuators of microrobots inside a microfluidic chip the resulting microrobots were able to perform desired functions, like moving or grasping. This work may help real-

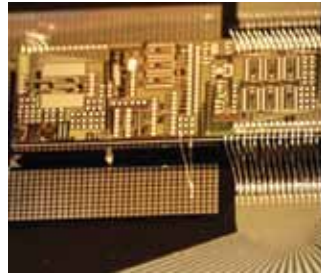
ize the vision of microsurgery performed by autonomous robots. Soft microrobot structures are produced by shining laser light to harden a photo-inducible biocompatible hydrogel, poly (ethylene glycol) diacrylate.

Neuromorphic skin to continuously monitor health

Wearable technology can be used as precision medical sensors designed to perform health monitoring and diagnosis. Health diagnosis requires continuous monitoring as the measurements could differ depending on the person's age, medical history, and other factors. Researchers have adapted the neuromorphic computing technology to create a device that could mimic the operation of the brain and train itself using past data sets. They designed a skin-like neuromorphic 'chip' consisting of a thin film of a plastic semiconductor combined with stretchable gold nanowire electrodes. Even when stretched to twice its normal size, the device functioned as planned without formation of any cracks. They developed an AI and trained it to distinguish healthy electrocardiogram (ECG) signals from four different signals indicating health problems. After training, the device was more than 95% effective at correctly identifying the ECG signals.

Optical chip trains artificial intelligence

The training of artificial intelligence (AI) systems costs a significant amount of energy and carbon footprint. Even a completely developed AI demands high computational power. With the introduction of photonics-integrated chips there seems to be a way to deliver higher computing perfor-



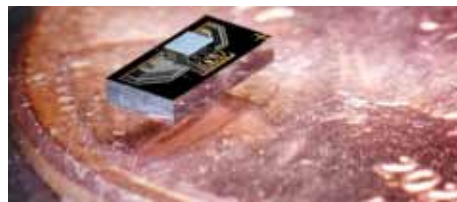
A picture of the chip used for this work (Credit: The George Washington University/Queens University)

mance. Researchers from various institutions have come together and developed an optical chip that can train machine learning hardware. Machine learning is a two-step procedure. In the first step the data is used to train the system and then there is a performance test of the AI system. Based on the re-

sults and the errors observed, the hardware is reconfigured for the second training cycle. Thus far, photonic chips have only demonstrated an ability to classify and infer information from data. Now, researchers have made it possible to speed up the training step itself.

A chip combo for ultrahigh speed data transmission

Researchers believe computation speed is becoming a "bottleneck, especially under stringent energy constraints." At Caltech and University of Southampton in England they have



The chip sandwich: an electronics chip (the smaller chip on the top) integrated with a photonics chip, sitting atop a penny for scale (Credit: Arian Hashemi Talkhooncheh)

designed an electronic chip integrated with a photonic chip (uses light to transfer data), which can transmit data at ultrahigh speed.

This device can be used for high-grade server rooms and data centres. As data processing is mostly done on electronics and the transmission does not necessarily require electronics, photonics (such as fibre-optic communication) can be used for data transmission. The resulting optimised interface between the two chips allows them to transmit 100 gigabits of data per second while producing just 2.4 pico-Joules per transmitted bit.

Brain-machine interface improves significantly

Researchers at the University of Texas have developed a system that takes direct inputs from the human brain through brain machine interface (BMI) and uses it as commands to drive the wheelchair. Researchers recruited three tetraplegic people and trained them for 2-5 months. Users wore a skullcap that detected their brain activities through electroencephalography (EEG), which were converted to mechanical commands for the wheelchairs via a brain-